Application Page Response Times

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To fully leverage the SPLUNK Monitoring Platform, business desired a means to both gauge and assess the V3 Application AIM and Billing Page Reponse Times. To elaborate, a given page response time represents a duration or differential in time, a value that will be generated and represented by time to "**serve**" the page and **not** the values taken by the server to generate the page or perform actions against the page. It would be this action that would be monitored by SPLUNK to determine Page Responsiveness.

In working with business, our development team was charged with devising SPLUNK Monitoring for the following V3 Application Page's (Requirements *Email attached for reference at the bottom of this page*):

- Reallocation
- Imports
    - Mass Updates
    - Positive Pay files
- Invoice Line items and all associated transactions within the recon process
    - Positive Pay file allocation
- All ESS screens
    - Load time when user first logs on and navigates to each screen
    - My Invoices tab
        - Positive Pay file imports
        - Invoice Reconciliation
        - Make Payment Wizard
    - My Associates tab
        - Editing / change Payor details
- Suspense Dashboards
- Workflows
- Tools>>Reports

With the requirements in hand, our team's first task was to drive out a Proof Of Concept(POC) and configure the SPLUNK Platform to recognize an Application Page Response Time.

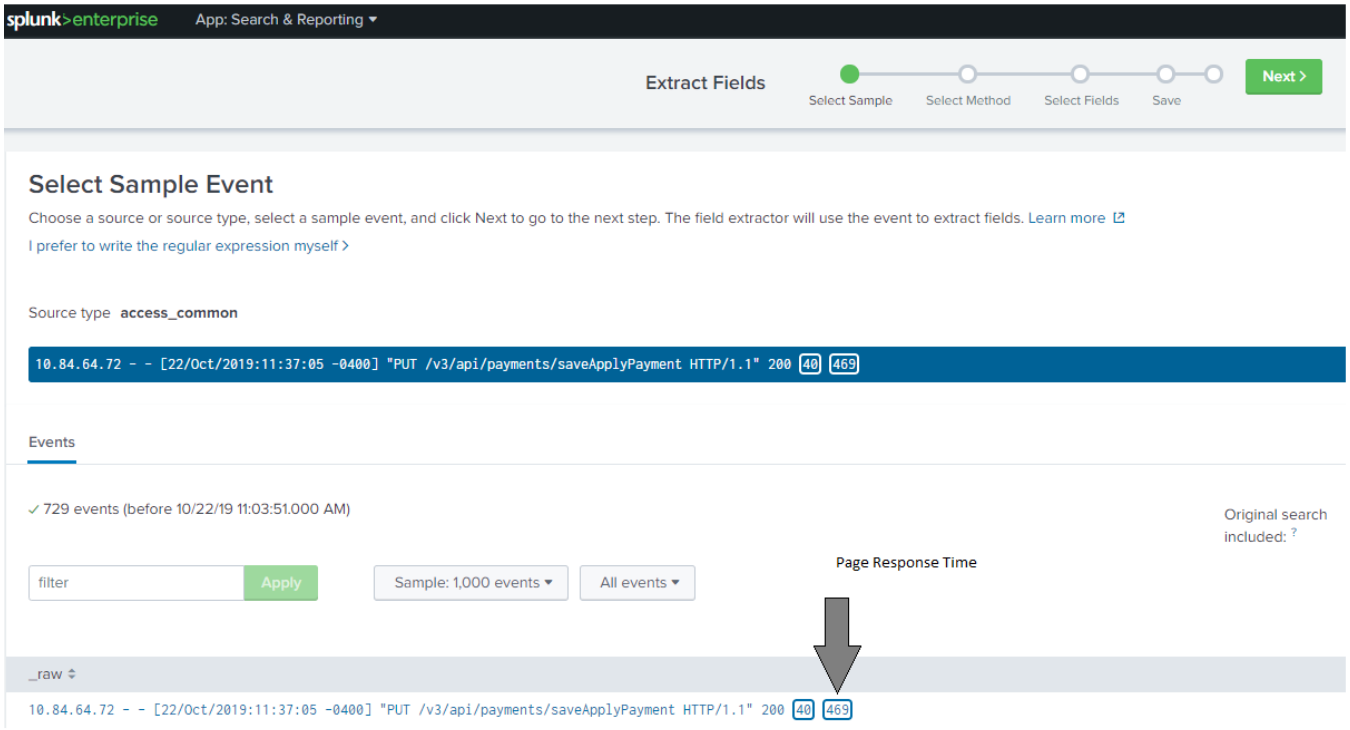
> It is presumed that the intended reader is of a technical mind and has a reasonable understanding of the SPLUNK Monitoring Platform. Please reference the following materials if otherwise;
>
> SPLUNK Reference Material: https://confluence.allstate.com/display/ADT/SPLUNK
>
> SPLUNK Example Query Strings: https://confluence.allstate.com/pages/viewpage.action?pageId=61947669

In order to do this, we, as developers must first write and execute a SPLUNK Query against the Web Application's **Access Common Log**, to retrieve log entries specific to the Application's **URI Path**. It is here we will find the **Page Response Time**.

We collaborated with the **AB Infrastructure and Platform Support Team** to configure the Application Servers. This was vitally crucial because such configuration would allow the application to record/write page response times to the Access Common Log.

Once the configuration was complete, and page response times were validated, within the Access Common Log, we performed the "**Extract Field**" Action, within the SPLUNK's Framework. Referencing the Screen shot below, the Grey Arrow depicts where in the **URI Path** the Response Time co-exists. By performing such an **Extraction** allows SPLUNK to interpret all URI Path Patterns, with basis upon Regular Expressions, in order to extract it's value into a **SPLUNK Field Declared (response_time)**.

Having performed those steps, our remaining hurdles were to identify all pages associated to those areas business desired monitored as well as construct SPLUNK Queries, leveraging Statistical Functionality, so that such metrics could later be published within SPLUNK Dashboards.



Identifying Page Names within the V3 Billing Application proved cumbersome at times. Traditionally, web application's 20 years ago, displayed the name of the page within the URI/URL Path. That is no longer the case today. Industry has taken on a greater level of obfuscation to enable or allow for a better security design implementation. It is felt by doing so, thwarts hacking or phishing attempts or at least minimizes them. In working with business and Arunbabu Sureshkumar's Test Team, we were able to delegate Page Uri Path Callouts within the Access Common Log. By having testers navigate to specific application areas, during a set time-frame, allowed us to adequately identify the pages desired for monitoring. Those results can be found below within the table:

Application Operation	Done By	Time In (IST)	Time Out (IST)	Time (in mins)
Reallocation	argab	13:30	13:31	1
Imports				
Mass Updates	argab	13:33	13:35	2
Positive Pay files	mthbe	14:26	14:30	4
Invoice Line items and all associated transactions within the recon process	mthbe		14:30	1
Positive Pay file allocation	mthbe		14:30	1
All ESS screens				
Load time when user first logs on and navigates to each screen	rnigc	13:44	13:44	1
Login page	rnigc	13:44	13:45	1
Home page	rnigc	13:45	13:45	1
Profile page	rnigc	13:45	13:46	1

Account summary page	rnigc	13:46	13:46	1
Invoices page	rnigc	13:46	13:46	1
Invoice Reconciliation	rnigc	15:02	15:02	1
Associate page	rnigc	13:47	13:48	1
Editing / change Payor details	rnigc	13:48	13:49	1
Forms page - export and form rendering verified	rnigc	13:49	13:50	1
Users page	rnigc	13:50	13:51	1
Communication page	rnigc	13:51	13:51	1
Make one time payment page	rnigc	13:52	13:53	1
Suspense Dashboards - Dashboard Load and xl download time verified	argab	13:54	13:56	2
Workflows	argab	13:56	13:59	3
Tools>>Reports - Report download scheduled	argab	13:59	14:01	1

The reader probably finds themselves asking, ok, fine, we went through all of this trouble what does the end result look like. As suggested previously, in collaborating with many colleagues, the SPLUNK Team, Infrastructure Team, Test Team, and Business we were able to construct the necessary SPLUNK Queries, leveraging delivered Statistical Modules, all while exporting them to panels which were eventually injected into a SPLUNK Dashboard. What follows is a screen shot of what those Page Response Times look like in our Performance Environment in addition to a Legend dictating Color Coded Definitions:

Currently the Page Response Time SPLUNK Dashboard has been broken out by the following sections:

(1) All Pages Avg Response Time

(2) All Pages Success Rate

(3) Max Avg Response Time by Page

(4) Max Response Time by Page

(5) Avg Response Time by Page

(6) Max Response Time by Page

(7) Count By Event Type

(8) Individual Page Metrics

SPLUNK - V3 APP Billing - Page Response Time Dashboard

V3 Prototype Dashboard - All Hosts

Edit | Export | ...

Time Picker

Week to date | Hide Filters

All Pages Avg Response Time

0.0 s

All Pages Success Rate

100.00 %

Max Avg Response Time By Page

0.0 s

Max Response Time By Page

2.1 s

Avg Response Time By Page (access_common)

uri_query	count	Avg Response	AvgTimeTrend
pageName=Datasheet:GUW	83	0.0 s	
pageName=Workflow:WorkManager	53	0.0 s	
pageName=SEARCH	125	0.0 s	
pageName=workflow:WorkManagerPage	51	0.0 s	
pageName=JasperReporting_ReportSetupPage	2	0.0 s	
pageName=AdminPages_DependentAffiliateTypesPage	1	0.0 s	
pageName=WorldRules:SystemSettingsPage	3	0.0 s	
pageName=World:Workflows	651	0.0 s	
pageName=CrmRules:CopyOppsOpportunity	1	0.0 s	
pageName=widget:EditControl	202	0.0 s	

« Prev 1 2 3 4 5 6 7 8 9 10 Next »

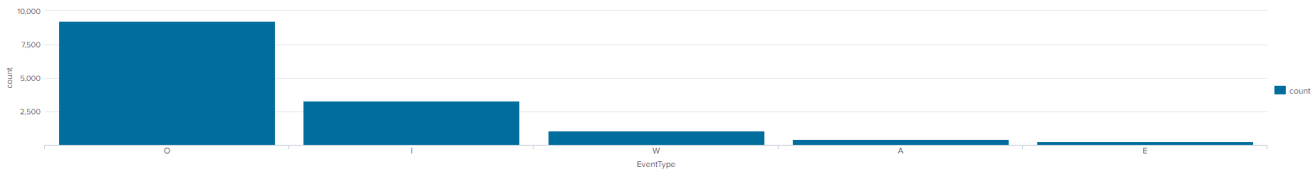
Max Response Time by Page (access_common)

uri_query	count	Max Response Time	MaxTimeTrend
pageName=Crm:Opportunity	1429	2.1 s	
pageName=Crm:OpportunityDocs	1429	2.1 s	
pageName=Crm:OpportunityWorkflow	1429	1.8 s	
pageName=Crm:OpportunityOffering	1429	1.8 s	
pageName=ESS:Documents	919	1.7 s	
pageName=Datasheet:GUW	83	1.5 s	
pageName=Datasheet:DS_SOLD_CASE_ATTRIBUTES	1427	1.4 s	
pageName=Datasheet:DS_DETAILS	1427	1.4 s	
pageName=Datasheet:DS_MARKETING_02	1427	1.4 s	
pageName=Datasheet:DS_PRODUCTS	1427	1.4 s	

« Prev 1 2 3 4 5 6 7 8 9 10 Next »

3m ago

Count by EventType (System Out)



Page: Reallocation	Page: Mass Update	Page: Positive Pay Files	Page: Invoice Line Items
0.0 s	0.0 s	0.0 s	0.0 s

Page: Home Page	Page: Login Page	Page: Profile	Page: Login and Navigate All Screens	Page: Account Summary
0.0 s	0.0 s	0.0 s	0.0 s	0.0 s

Page: Invoices	Page: Invoice Reconciliation	Page: My Associates	Page: Edit Payor Details
0.0 s	0.0 s	0.0 s	0.0 s

Page: Forms - Export and Form Rendering Verified	Page: Users	Page: Communication	Page: Payment Wizard
0.0 s	0.0 s	0.0 s	0.0 s

LEGEND - Color Range Definitions

- The following Color Scale (Green => Red) has been chosen to visually depict (Success => Failure), with Grey being a Default Color for no value found

*** Currently the Page Response Time Dashboard has not been published to Production. We hope this will happen within the near term. ***

~For Reference

Requirements Email :	RE_ AIM a...oring.msg	**Response Time Value Definition:**	RE_ V3 B...ECC`.msg	**Response Time Results Email:**	RE_ AIM a...oring.msg	**SPLUNK Dashboard Source:**	V3 Splunk ...ed Up).txt